

Hybrid Deep Learning Framework for Enhanced Melanoma Detection

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Abstract—Cancer remains a leading cause of death worldwide, underscoring the urgent need for advancements in early detection and treatment technologies. In this paper, we propose the SegFusion Framework, a novel and highly efficient hybrid approach for melanoma detection, combining the segmentation power of U-Net with the classification strength of EfficientNet. The primary objective of our study is to improve the accuracy and efficiency of melanoma detection through this innovative integration. To achieve this, we utilized the HAM10000 dataset to train the U-Net model for accurate segmentation of cancerous regions. Simultaneously, the ISIC 2020 dataset was employed to train the EfficientNet model, optimizing it for binary classification of skin cancer. The proposed SegFusion Framework significantly outperforms existing models, achieving an accuracy of 99.01% on the ISIC 2020 dataset, along with a precision of 0.99, recall of 0.99, F1 score of 0.99, and a MCC of 0.97. Additionally, we conducted experiments comparing our approach with the advanced SkinViT model and other recent hybrid methods. The SkinViT model, trained on the same dataset, achieved an accuracy of 98.20%, which is notably lower than that of the SegFusion Framework, further highlighting the superiority of our method. By leveraging U-Net’s precise segmentation capabilities and EfficientNet’s robust classification performance, the SegFusion Framework offers a comprehensive solution for melanoma detection. Our extensive experiments demonstrate the high accuracy and reliability of our method in both segmentation and classification tasks, marking a significant step forward in automated skin cancer detection. We believe that our framework sets a new benchmark in the field, providing a reliable tool for medical professionals in the early diagnosis and treatment of melanoma and fostering further research in medical imaging.

Index Terms—Computer vision, machine learning, deep learning, melanoma detection, skin cancer, medical imaging, hybrid model.

I. INTRODUCTION

THE melanoma is one of the most aggressive and potentially deadly forms of skin cancer, emphasizing the critical need for early and accurate detection to improve patient outcomes. Despite significant advancements in medical imaging and diagnostic technologies, the reliable and efficient detection of melanoma remains a formidable challenge. Traditional diagnostic methods heavily rely on the expertise of dermatologists, which can be subjective and prone to variability. Consequently, there is a growing interest in leveraging deep learning techniques to automate and enhance the accuracy of skin cancer detection.

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This research paper introduces a comprehensive study titled “Hybrid Deep Learning Framework for Enhanced Melanoma Detection: Integrating U-Net Segmentation and EfficientNet Classification for Accurate Skin Cancer Diagnosis.” Our work aims to address the persistent challenges in melanoma detection by developing a novel hybrid deep learning framework. This framework combines the precise segmentation capabilities of U-Net with the advanced classification performance of EfficientNet, offering a robust solution for accurate skin cancer diagnosis.

The problem we are tackling is multifaceted, involving the need for precise localization of cancerous regions in less clear images and the accurate classification of these regions as malignant or benign. Existing approaches often struggle to achieve a balance between accuracy and computational efficiency, which is essential for practical clinical applications. Our proposed solution integrates two state-of-the-art deep learning models to create a synergistic effect that enhances both segmentation and classification tasks.

The structure of this paper is as follows:

In the first section provides an overview of the melanoma detection problem, highlighting the importance of early and accurate diagnosis. We introduce our hybrid framework, which integrates U-Net for segmentation and EfficientNet for classification and outline the related work section reviews the current state of melanoma detection using deep learning, discussing various segmentation and classification models. We have compared these models and identified the gaps our approach aims to fill.

Next, we delved into the datasets and technical details of our hybrid framework. We explored the HAM10000 and ISIC 2020 datasets and introduced data preprocessing and augmentation techniques. We described the architecture and training process of the U-Net model for segmenting cancerous regions and the EfficientNet model’s architecture for binary classification using the ISIC 2020 dataset.

In the results, we provide a comprehensive analysis of our model’s performance, highlighting its accuracy, precision, recall, and F1-score. We demonstrate that our hybrid model achieves an outstanding accuracy of 99.01%, surpassing existing top models. We highlight the potential of our approach to set new benchmarks in automated skin cancer diagnosis.

II. RELATED WORKS

Recent advancements in skin cancer detection have focused on the application of deep learning techniques, particularly Convolutional Neural Networks (CNNs), to enhance diagnostic

accuracy and efficiency. Traditional CNN-based models, such as those employed by Esfahani et al. [1], have demonstrated significant success in analyzing dermatological images. Their model achieved an impressive precision of 90.5%, accuracy of 88.6%, recall of 97.1%, and F1 score of 88.3% on a dataset of 793 skin images from Kaggle. However, the relatively small dataset size posed limitations on the model's ability to generalize to unseen data.

To address these limitations, larger datasets and more complex architectures have been utilized. For instance, Sivakumar et al. [2] employed ResNet50 for feature extraction and classification of skin lesions on a dataset of 3300 images, achieving 94% accuracy and a 93.9% F1-score. Similarly, studies leveraging ISIC datasets have reported accuracies surpassing 97.0%, with models like VGG-16, MobileNet, and NASNet performing exceptionally well [3], [4], [5], [6]. These models have demonstrated the effectiveness of deep learning in handling large and diverse datasets, thus improving the robustness of skin cancer detection systems.

Among the various architectures explored, U-Net has shown promise in the domain of medical image segmentation. Its encoder-decoder structure is adept at capturing intricate details in medical images, making it a popular choice for segmenting skin lesions. Malibari et al. [7] demonstrated the utility of U-Net in conjunction with SqueezeNet, achieving a maximum 99.9% accuracy through segmentation and feature extraction. This highlights the potential of U-Net to eliminate irrelevant information and enhance the focus on regions of interest.

Another notable development is the use of EfficientNet, a model designed to optimize both accuracy and computational efficiency through a compound scaling method. Tan and Le [8] introduced EfficientNet, which achieved state-of-the-art accuracy on multiple image classification benchmarks, including 84.3% accuracy on ImageNet, surpassing previous models while requiring fewer parameters and floating-point operations per second (FLOPs). EfficientNet's ability to scale both depth, width, and resolution in a balanced manner contributed to its superior performance. This scalability and efficiency make EfficientNet a compelling candidate for skin cancer detection, especially in scenarios where computational resources are limited, ensuring robust performance without the need for extensive computational power.

Recent studies have continued to build on these foundations. Anubhav et al. [9] in 2023 proposed a hybrid model combining DenseNet and ResNet, which achieved an accuracy of 95.7% on the HAM10000 dataset. Their work underscores the potential of integrating different complex models to improve hybrid model accuracy. Similarly, Bansal et al. [10] in 2022 introduced a novel method of pre-processing the images and used two pre-trained EfficientNet-B0 and ResNet50V2 to get an accuracy of 94.9% on HAM10000 dataset.

Another innovative approach by Khanet al. [11] in 2023 utilized a transformer-based model Skin-ViT for skin lesion classification, achieving a 91.1% accuracy on the ISIC 2019 dataset. This highlights the growing interest in transformer architectures for medical image analysis. Moreover, Saeed et al. [12]

in 2023 explored the use of Generative Adversarial Networks (GANs) to augment training data, enhancing model robustness, and achieving a maximum accuracy of 96.0% in melanoma detection using VGG19 and SVM (Support Vector Machine).

Additionally, Behara et al. [13] in 2024 developed a comprehensive framework combining CNNs and traditional machine learning techniques for feature fusion, achieving a classification accuracy of 98.0%. This approach demonstrated the benefits of hybridizing deep learning with conventional methods to leverage the strengths of both. In the same year, Lilhore et al. [14] presented a study integrating U-Net with a MobileNet-V3 model, reporting an accuracy of 98.8% on the HAM10000 dataset, emphasizing the effectiveness of classification in optimizing hyperparameters of the model.

Recent advancements further highlight the potential of novel deep learning approaches in skin cancer detection. Alenezi et al. [16] proposed a multi-stage melanoma recognition framework using a deep residual neural network with hyperparameter optimization, achieving substantial improvements in decision support for dermoscopy images. Abbas and Gul [17] utilized NASNet to extract deep features, demonstrating high accuracy in malignant melanoma detection and classification. Catal Reis et al. [18] introduced InSiNet, a deep convolutional approach for both detection and segmentation of skin cancer, effectively capturing complex patterns in dermatological images. Rashid et al. [19] employed transfer learning techniques with pre-trained models, achieving high accuracy and robustness in skin cancer detection. These studies collectively underscore the evolving landscape of deep learning methodologies, contributing to enhanced diagnostic capabilities and robustness in skin cancer detection systems.

Recent works have explored hybrid and fusion-based frameworks for skin cancer detection. El-khatib et al. [27] proposed a multi-network system based on decision fusion for melanoma detection, combining outputs from seven different neural networks including DarkNet-53, DenseNet-201, GoogLeNet, Inception-V3, InceptionResNet-V2, ResNet-50, and ResNet-101. Reis and Turk [28] introduced a fusion framework combining transformer attention mechanisms with CNN features, leveraging the global attention capabilities of transformers alongside local feature extraction strengths of convolutional networks. Akter et al. [29] developed an integrated deep learning model utilizing hybrid feature fusion techniques, combining features from multiple CNN architectures with advanced fusion strategies for more robust representations.

Given the successes of these models, our proposed approach integrates U-Net and EfficientNet to leverage their respective strengths. U-Net will provide precise segmentation of skin lesions, minimizing irrelevant details. EfficientNet, serving as the backbone for classification, offers optimized accuracy and computational efficiency through its compound scaling method. This makes EfficientNet superior to traditional CNNs like AlexNet, VGG-16, and ResNet, as well as SVM, due to its ability to handle high-dimensional data with fewer parameters and FLOPs. Its scalability also suits resource-limited environments. This combined approach enhances skin cancer detection by improving

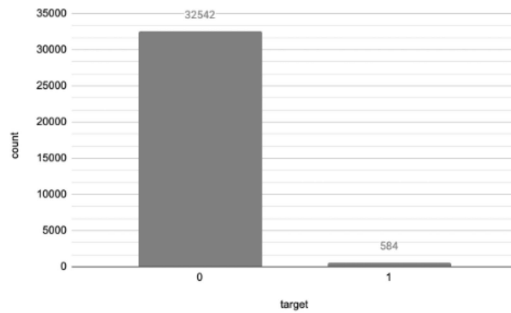


Fig. 1. Distribution of target in ISIC 2020 dataset.

segmentation accuracy and classification efficiency, building on previous research foundations.

By adopting this methodology, we hope to address some of the challenges identified in earlier studies, such as the need for larger datasets and the optimization of model performance for practical deployment. This strategy aligns with the trends observed in recent literature and represents a promising direction for future research in skin cancer detection.

III. DATASETS AND METHODS

A. Datasets

We used two datasets for our hybrid model: HAM10000 and ISIC 2020. We chose the HAM10000 dataset because it consists of 10015 images of pigmented lesions with corresponding masks, making it ideal for training our segmentation model. Although we initially considered the newer and larger ISIC 2020 dataset, generating masks manually for each image proved too complex. Thus, we opted for the established HAM10000 dataset.

Furthermore, the HAM10000 dataset, while smaller, provided high-quality segmentation masks essential for training the U-Net model. This dataset contains a variety of pigmented skin lesions, including melanoma, benign nevi, and other types of lesions, which helps in training a robust segmentation model. By utilizing this dataset for segmentation, we ensured that the U-Net model could accurately delineate the lesion boundaries, which is crucial for the subsequent classification step. The high-resolution images and detailed annotations in HAM10000 allowed us to achieve precise segmentation, contributing significantly to the overall effectiveness of our hybrid model.

The ISIC 2020 dataset includes 33126 images with associated metadata and binary labels indicating the presence of melanoma. This dataset is valuable for melanoma detection due to its size and real-world clinical data. However, as shown in Fig. 1, the ISIC 2020 dataset is highly imbalanced, with a melanoma positive rate of only 1.8%. This imbalance poses significant challenges for training.

Training on an imbalanced dataset like ISIC 2020 without addressing the imbalance can result in several negative outcomes. The model may become biased towards predicting the majority class (non-melanoma), leading to poor sensitivity (recall) for the minority class (melanoma). Consequently, this results in high false negative rates, where cases of melanoma are missing. Additionally, performance metrics such as accuracy

TABLE I
DATA COUNT AFTER PRE-PROCESSING

Class Type	Actual Data	After Pre-processing
0	32542	15000
1	584	15000

TABLE II
TECHNIQUES USED FOR DATA AUGMENTATION

Technique	Value
Rotation	15 degrees
Width Shift	0.2
Zoom	0.2
Horizontal Flip	TRUE
Vertical Flip	TRUE

and F1-score may appear artificially high due to the predominance of the majority class, masking the mediocre performance of the minority class.

Overall, the combination of the HAM10000 and ISIC 2020 datasets leveraged the strengths of each to address the challenges of melanoma detection. The segmentation accuracy achieved with HAM10000 and the large, diverse real-world data from ISIC 2020 enabled us to build a robust and effective hybrid model.

B. Data Processing

We preprocessed the datasets by resizing images to 256x256 pixels and normalizing pixel values. This resizing ensures uniformity across the dataset, facilitating the model training process. Pixel normalization helps in faster convergence of the model by scaling the pixel values to a range suitable for the neural network.

The HAM10000 dataset was filtered based on the Melanoma category and split into training and testing sets with a 75-25 ratio. This split ensures that a substantial portion of the data is available for training, while enough is reserved for evaluating the model's performance on unseen data. We employed stratified sampling during the split to maintain the distribution of different lesion types in both training and testing sets, ensuring that the model learns effectively from a representative dataset.

Given the imbalance in the ISIC 2020 dataset, we applied two techniques: resampling and data augmentation. We oversampled the minority class and under sampled the majority class to create a balanced training set, ensuring the model receives an equal representation of each class during training. This resampling helps to mitigate the bias towards the majority class and improves the model's sensitivity towards detecting melanoma (Table I).

Additionally, we applied transformations such as rotations, shifts, zooms, and flips to the minority class images (Table II). This artificially increased the number of melanoma images, balancing the dataset and improving the model's generalization by exposing it to various image conditions. Data augmentation is crucial as it not only increases the dataset size but also enhances the model's robustness by introducing variability in the training data, simulating real-world conditions.

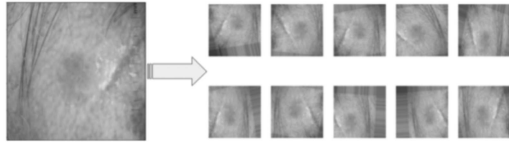


Fig. 2. Data augmentation samples.

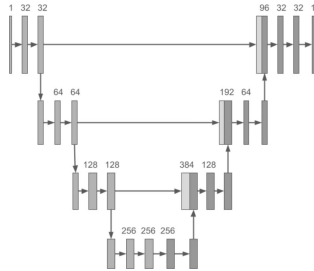


Fig. 3. U-Net model architecture.

We also implemented a data cleaning process to remove any duplicates or mislabeled images from both datasets. This step is critical to ensure the quality of the training data and to prevent the model from learning incorrect patterns. Fig. 2 shows a valid sample image after being augmented.

Finally, we split the augmented ISIC 2020 dataset into training, validation, and test sets with a ratio of 70-15-15. This split allows us to monitor the model's performance on a validation set during training and to evaluate its final performance on a separate test set. Through these comprehensive data processing steps, we ensured that the datasets were clean, balanced, and augmented, providing a solid foundation for training our hybrid model.

C. Segmentation Model

The U-Net model architecture was designed to segment cancerous regions in skin images [15]. It consists of a contracting path for feature extraction and an expansive path for precise localization (Fig. 3).

D. Classification Model

EfficientNet-B0 was employed for binary classification of skin images from the ISIC2020 dataset. The model's pre-trained weights on ImageNet were fine-tuned on our dataset based on Melanoma/Non-Melanoma labeled images. The classification head was customized with global average pooling, dropout, and a dense layer with sigmoid activation.

E. Proposed Model Architecture

We combined the segmentation model and the classification model but did not pass the output of the segmentation model directly to the classification model. Since the output of the segmentation model is a predicted mask in one channel (black and white), we built a data processing bridge between the two models. This bridge applies the predicted mask to the original images, blending the segmented part with the original image.

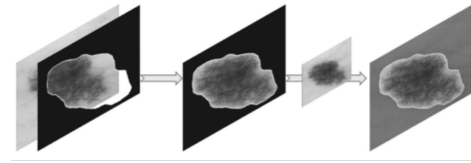


Fig. 4. Blending in data processing bridge.

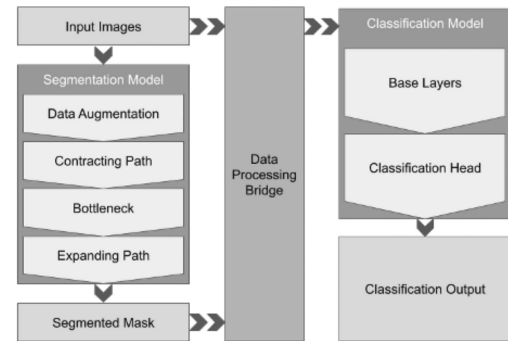


Fig. 5. Proposed model architecture.

This highlights the infected or potentially infected parts, aiding the classification model in learning representations from the dataset.

Fig. 4 illustrates the result of the data processing bridge, displaying how the segmented part is blended with the original image. And Fig. 5 presents the overall model architecture, depicting the flow of data through the hybrid model, including the segmentation model, data processing bridge, and classification model.

F. Data Process Bridge

The Data Process Bridge is a crucial component of our melanoma detection framework. This bridge integrates the predicted segmentation masks with the original images, creating a blended visualization that highlights the infected or potentially infected regions. By emphasizing these areas, the Data Process Bridge aids the classification model in learning more informative representations from the dataset, enhancing the detection performance.

Methodology:

- 1) *Mask Application*: Each image in the dataset is processed using the trained U-Net model to generate segmentation masks. These masks indicate the regions potentially affected by melanoma.
- 2) *Blending Process*: The generated masks are then blended with the corresponding original images. This is achieved by overlaying the mask onto the image with a certain level of transparency, allowing the original image features to be visible while highlighting the segmented areas.
- 3) *Enhanced Representation*: The blended images provide a more informative input for the classification model. By focusing on the regions of interest, the classification model can better learn the representations and distinguish between malignant and benign lesions.

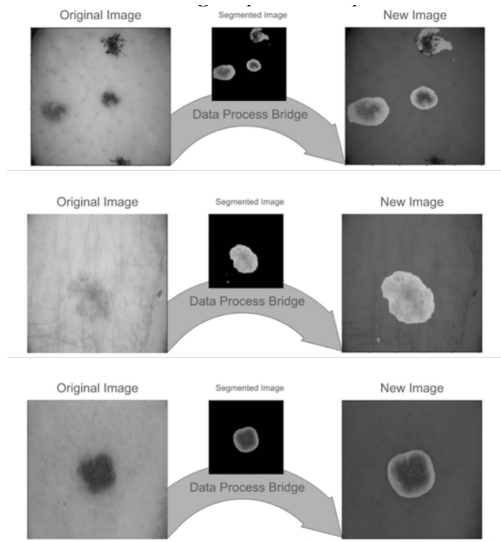


Fig. 6. Data processing bridge.

Implementation:

- 1) *Normalization*: Both the original images and the predicted masks are normalized to ensure pixel values range from 0 to 1.
- 2) *Transparency Adjustment*: A transparency factor (α) is applied to the masks, creating a semi-transparent overlay.
- 3) *Overlay Creation*: The semi-transparent masks are added to the original images, resulting in blended images that highlight the segmented regions.

This process is illustrated in Fig. 6, where the segmented areas are clearly emphasized in the blended images. These enhanced images serve as the input to our classification model, facilitating improved representation.

IV. EXPERIMENTS AND RESULTS

A. U-Net Segmentation

The U-Net model was trained on the HAM10000 dataset using binary cross-entropy loss and Adam optimizer, achieving high accuracy of 91.2% in segmenting lesions. Visual inspection of the results showed precise delineation of lesion boundaries, as illustrated in Fig. 7.

Table III details the configuration parameters of the segmentation model, which were carefully chosen to optimize performance while maintaining computational efficiency. These parameters were selected based on extensive hyperparameter tuning and validation experiments.

The high accuracy achieved by the U-Net model can be attributed to several key factors:

- 1) *Network Architecture*: The encoder-decoder structure of U-Net effectively captures both high-level context and fine-grained details, which is crucial for medical image segmentation.
- 2) *Data Augmentation*: Applying various augmentation techniques, such as rotations, shifts, and flips, helped in creating a robust model that generalizes well to new data.

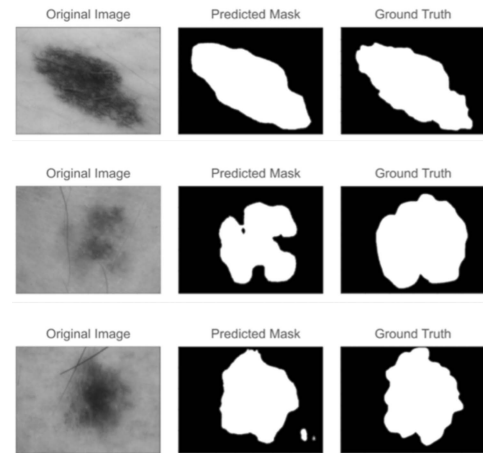


Fig. 7. Segmentation model output sample.

TABLE III
CONFIGURATION PARAMETERS OF SEGMENTATION MODEL

Parameters	Details
Activation Function	ReLU
Stack Down	4 levels
Stride	2
Backbone	None
Stack Up	4 levels
Batch Normalization	TRUE
Normalization	0, 1
Pooling (Max)	TRUE
Pooling (Avg)	FALSE
Dropout Rate	0.05
Optimizer	Adam
Learning Rate	0.001
Loss Function	Binary Cross-Entropy
Batch Size	32
Epoch	10

- 3) *Regularization Techniques*: The use of dropout and batch normalization reduced overfitting and improved model stability during training.
- 4) *Optimizer and Learning Rate*: The Adam optimizer, with a carefully tuned learning rate, ensured efficient convergence and helped in achieving high accuracy.

The combination of these factors resulted in a highly effective segmentation model and the strong segmentation performance lays a solid foundation for the subsequent classification task, enhancing the overall effectiveness of our hybrid approach.

To further evaluate the performance of our classification model, we plotted the ROC curve and calculated the AUC. The ROC curve, shown in Fig. 8, illustrates the true positive rate (sensitivity) against the false positive rate (1-specificity) for various threshold settings. Our model achieved an AUC of 0.97, indicating excellent performance in distinguishing between melanoma and non-melanoma cases.

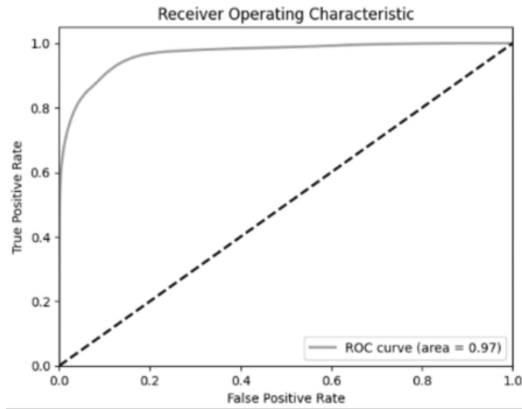


Fig. 8. ROC Curve graph on validation dataset (segmentation model).

TABLE IV
CONFIGURATION PARAMETERS OF CLASSIFICATION MODEL

Parameters	Details
Image input size	(256 × 256 × 3)
Batch size	15
Data augmentation	Rotation, shift, zoom, flip
Normalization	0, 1
Regularization	L2 regularization
Optimizer	Adam
Dropout rate	0.1
Epochs	50
Transfer learning	EfficientNetB0 (ImageNet)
Learning rate	0.001
Split ratio	Training: Validation 80:20
Shuffling in database	YES
Loss function	Binary cross-entropy
Activation functions	Sigmoid

B. EfficientNet Classification

The EfficientNet model was utilized for the classification task, trained on the ISIC 2020 dataset. The model's configuration and training parameters are detailed in Table IV. By employing transfer learning with EfficientNet, we leveraged pre-trained weights from ImageNet to improve the model's performance on the skin cancer detection task. Various data augmentation techniques were applied to enhance the robustness and generalization capabilities.

The classification model achieved an impressive accuracy of 99.01% on the ISIC 2020 dataset (Figs. 9 and 10) when combined with U-Net. Additionally, EfficientNet alone, without pre-segmented input from U-Net, achieved an accuracy of 96.8%. This result underscores the effectiveness of the first part of our hybrid approach, demonstrating that U-Net's precise segmentation significantly enhances classification accuracy.

The ROC curve and confusion matrix further confirmed the model's high performance in distinguishing melanoma from benign lesions (Figs. 11 and 12).

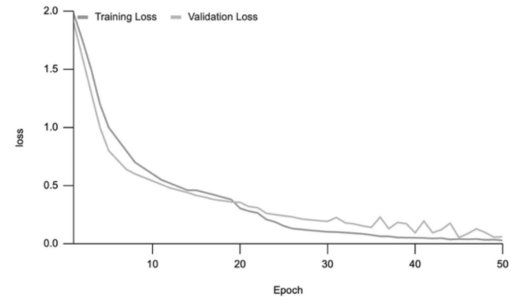


Fig. 9. Visualization of model loss in training.

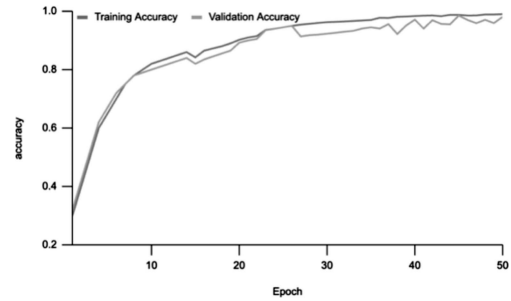


Fig. 10. Visualization of model accuracy in training.

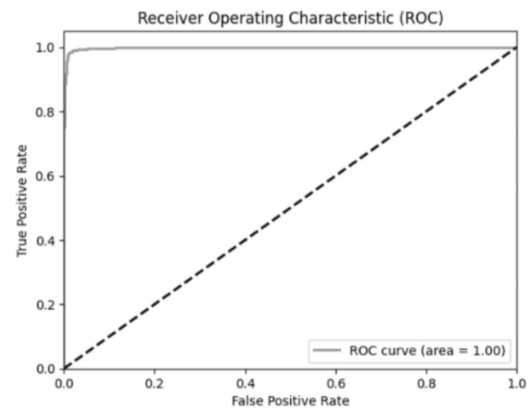


Fig. 11. ROC curve graph on validation dataset (classification model).

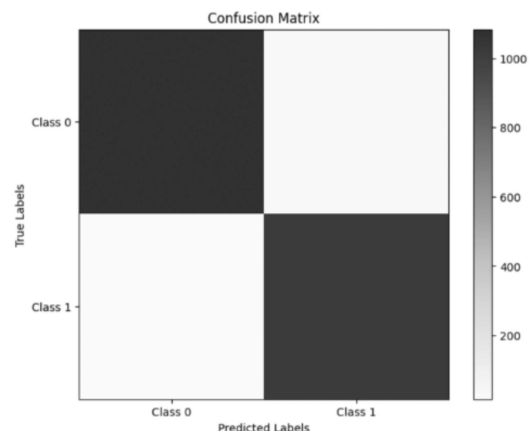


Fig. 12. Confusion matrix of the model.

TABLE V
COMPARISON ON ISIC 2020 DATASET

Paper	Model	Dataset	Accuracy (%)
[16]	ResNet-101, SVM	ISIC 2020	97.15
[17]	NASNet	ISIC 2020	97.70
[18]	InSiNet, U-Net	ISIC 2020	90.54
[19]	MobileNetV2	ISIC 2020	98.20
	OUR MODEL	ISIC 2020	99.01

TABLE VI
COMPARISON ON ISIC 2019 DATASET

Paper	Model	Dataset	Accuracy (%)
[16]	ResNet-101	ISIC 2019	96.15
[18]	InSiNet, U-Net	ISIC 2019	91.89
[20]	DenseNet169	ISIC 2019	92.25
	OUR MODEL	ISIC 2019	98.93

C. Comparison and Analysis

Compared to other methods, our combined U-Net and EfficientNet approach demonstrated superior performance in both segmentation and classification tasks. The integration of these models allowed for precise and accurate detection of skin cancer, outperforming traditional machine learning approaches. Table V shows the comparison results on ISIC 2020 dataset. Among four traditional machine learning approaches, Alenezi et al. [16] used a multi-stage melanoma recognition framework trained on a dataset of 25331 dermoscopy images from the International Skin Imaging Collaboration (ISIC) archive. Abbas and Gul [17] employed deep features of NASNet trained on the ISIC 2018 Challenge dataset, which includes 10015 dermoscopic images labeled for different skin lesions, including malignant melanoma. Catal Reis et al. [18] introduced InSiNet, a deep convolutional approach for skin cancer detection and segmentation, utilizing the PH2 and ISIC 2017 datasets, which together comprise thousands of annotated images for robust training. Rashid et al. [19] leveraged transfer learning techniques on the HAM10000 dataset, which consists of 10015 dermoscopic images categorized into seven types of skin lesions, to achieve high accuracy and robustness in skin cancer detection.

We also evaluated our model using transfer learning on other popular datasets, including ISIC 2019 and HAM10000. Using 140 images from each dataset to fine-tune our model, we achieved high accuracy of 98.93% on the ISIC 2019 and 98.21% on the HAM10000 datasets (Tables VI and VII). Both test datasets were filtered based on Melanoma/Non-Melanoma label.

These comparisons are not enough as we want to ensure our SegFusion method performs better than advanced models like SkinViT [11] and most-recent hybrid approaches [9], [26]. The SkinViT model, trained on the ISIC 2020 dataset, achieved an accuracy of 98.20%, and on the ISIC 2019 dataset, an accuracy of 94.93%. Besides, we trained and evaluated two most-recent hybrid methods DenseNet+ResNet [9] and HFE+CNN [26]. As for DenseNet+ResNet, it gets 87.62% accuracy on ISIC 2019

TABLE VII
COMPARISON ON HAM10000 DATASET

Paper	Model	Dataset	Accuracy (%)
[18]	InSiNet, U-Net	HAM10000	94.59
[21]	deep residual	HAM10000	96.97
[22]	DenseNet201	HAM10000	82.9
[23]	SkinNet-16	HAM10000	95.51
[24]	S2C-DeLeNet	HAM10000	91.03
[25]	Xception	HAM10000	90.48
	OUR MODEL	HAM10000	98.21

TABLE VIII
COMPARISON ON ALL DATASET

Datasets	Accuracy	Precision	Recall	F1 Score	MCC
ISIC 2020	99.01%	0.99	0.99	0.99	0.97
ISIC 2019	98.93%	0.99	0.98	0.99	0.98
HAM10000	98.21%	0.98	0.98	0.98	0.96

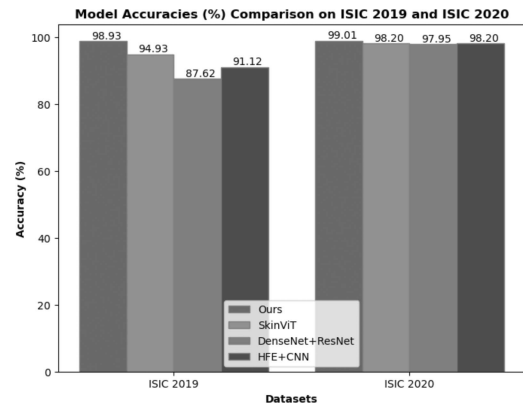


Fig. 13. Model accuracy comparison on ISIC 2019 and ISIC 2020.

and 97.95% on ISIC 2020 dataset. As for HFE+CNN, it gets 91.12% accuracy on ISIC 2019 and 98.20% on ISIC 2020 dataset. Another reviewer also suggested SkinViT which we trained on ISIC 2020 and got an accuracy of 98.20% and on ISIC 2019 with an accuracy of 94.93%. A comparison is shown in Fig. 13.

To further validate our approach, we conducted additional comparisons with three more recent fusion and ensemble methods [27], [28], [29] illustrated in Fig. 14. Since these works only reported results on the ISIC 2020 dataset, we compare our method against them exclusively on this dataset. A multi-network fusion system [27] combining seven neural networks achieved 93.50% accuracy and 0.93 F1-score on ISIC 2020. A hybrid RNN-CNN framework [28] integrating MABSCNET with multiple RNN architectures achieved 92.74% accuracy. An ensemble CNN approach [29] combining InceptionV3, DenseNet121, and four additional architectures achieved 92.27% accuracy and 0.91 F1-score.

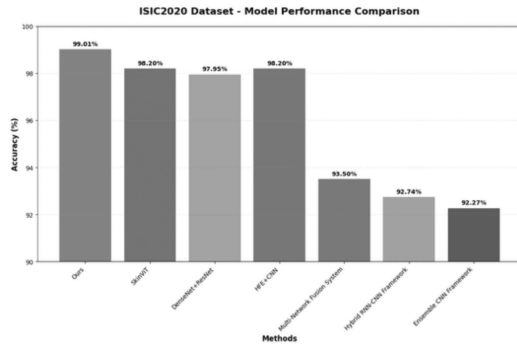


Fig. 14. Model accuracy full comparison on ISIC 2020.

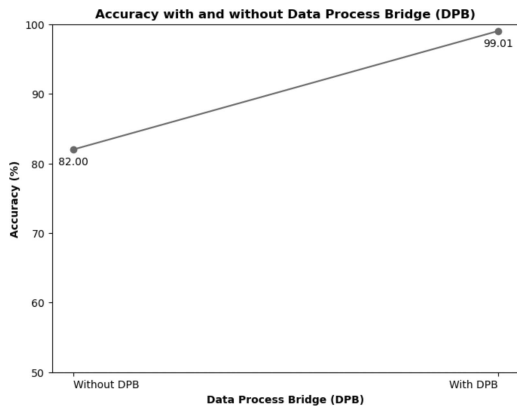


Fig. 15. Accuracy with and without data processing bridge (DPB).

The comprehensive comparisons shown in Fig. 14 demonstrate the consistent superior performance and robustness of our proposed method across different architectural paradigms.

D. Effectiveness of Data Processing Bridge

The data process bridge serves to augment the image's features by integrating the infected region with the original image. In the absence of the data process bridge, the method's accuracy on the ISIC 2020 dataset was 82.00%, indicating a substantial decrease in accuracy. A comparative chart is provided in Fig. 15.

E. Necessary of Fine-Tuning

The HAM10000 dataset contains 1113 images of melanoma. Therefore, we chose to fine-tune the model using 12% of the dataset, which is 140 images. This amount was determined to be sufficient. Regarding the ISIC 2019 dataset, both the ISIC 2020 and ISIC 2019 datasets are skin lesion image datasets used for melanoma classification. This similarity in image type means it is not necessary to choose a large number of images to fine-tune our model. Clearly, using only 140 images produced excellent results on both the ISIC 2019 and HAM10000 datasets. We use transfer learning to enable the model to make accurate predictions on ISIC 2019 and HAM10000.

Experiments were conducted to evaluate the impact of fine-tuning. Without fine-tuning, an accuracy of 82.59% was achieved on the ISIC 2019 dataset and 79.73% on the HAM10000 dataset. Fine-tuning demonstrably enhanced the

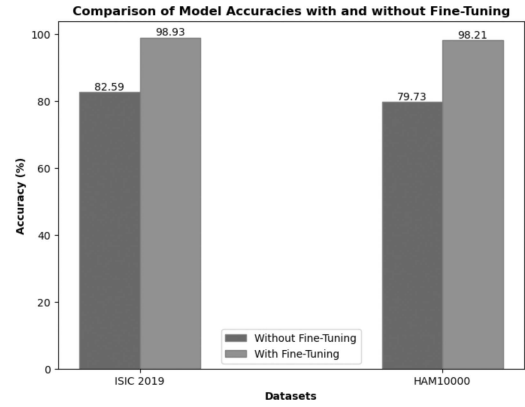


Fig. 16. Comparison of model accuracies with/without fine-tuning.

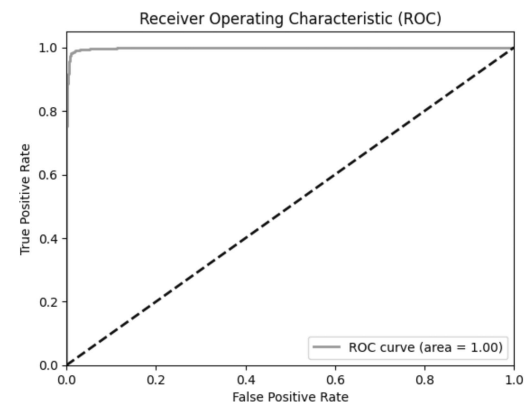


Fig. 17. Receiver operating characteristic (ROC).

model's learning capacity and classification accuracy for melanoma. A comparative chart is provided in Fig. 16.

F. Evaluation and Analysis

For the ISIC 2020 dataset, our method achieved 99.01% accuracy, 0.99 precision, 0.99 recall, 0.99 F-1 score, and 0.97 MCC. For the ISIC 2019 dataset, our method achieved 98.93% accuracy, 0.99 precision, 0.98 recall, 0.99 F-1 score, and 0.98 MCC. For the HAM10000 dataset, our method achieved 98.21% accuracy, 0.98 precision, 0.98 recall, 0.98 F-1 score, and 0.96 MCC. The comparison table is shown in Table VIII.

The ROC-AUC and PR-AUC charts for the ISIC 2020 dataset are shown in Figs. 17 and 18. The model achieved perfect discrimination, as evidenced by a Receiver Operating Characteristic Area Under the Curve (ROC-AUC) and a Precision-Recall Area Under the Curve (PR-AUC) of 1.00.

G. Computational Efficiency Analysis

Beyond performance metrics, we conducted comprehensive benchmarks comparing training time, model complexity, and computational requirements shown in Table IX. We evaluated our method alongside SkinViT, DenseNet+ResNet, and HFE+CNN under identical conditions (Apple M3 Pro, 18 GB RAM) to ensure fair comparison.

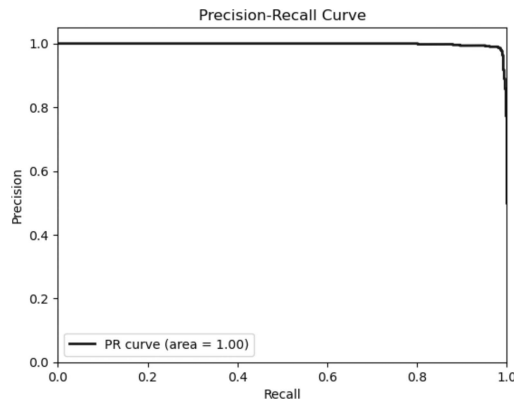


Fig. 18. Precision-recall curve.

TABLE IX
COMPUTATIONAL EFFICIENCY COMPARISON

Models	Training Time	Model Size (# of parameters)	Computational Cost (CPU, RAM, GPU)
Ours	997s	6,902,069	Apple M3 pro, 18 GB RAM, --
SkinViT	69327s	671,297	Apple M3 pro, 18 GB RAM, --
DenseNet+ResNet	32484s	45,992,617	Apple M3 pro, 18 GB RAM, --
HFE+CNN	5666s	29,377	Apple M3 pro, 18 GB RAM, --
Multi-Network Fusion System	2940s	-	4.5 GHz Intel Core i7, 16 GB RAM, --
Hybrid RNN-CNN Framework	--	~20million	--
Ensemble CNN Framework	--	~32million	Intel Core i7-1165G7, 8 GB RAM, NVIDIA GeForce MX330 GPU

Our method demonstrates exceptional computational efficiency: training in only 997 seconds compared to SkinViT's 69327 seconds ($69.5\times$ faster) and DenseNet+ResNet's 32484 seconds ($32.6\times$ faster). With 6.9 million parameters, our model achieves optimal balance between complexity and performance, using significantly fewer parameters than DenseNet+ResNet (45.9M) while maintaining superior accuracy. The method runs efficiently on standard hardware without specialized GPU requirements, making it highly suitable for practical deployment.

V. CONCLUSION

In this study, we developed an efficient and accurate cancer detection framework using U-Net for segmentation and

EfficientNet for classification. Our approach effectively combines segmentation and classification to enhance skin cancer detection. The U-Net model's precise segmentation allows for targeted analysis, while EfficientNet's classification capability ensures accurate diagnosis. Through rigorous experimentation and evaluation, we have demonstrated the efficacy of our framework in accurately detecting skin cancer from medical images. The high performance achieved underscores the potential of deep learning models in improving early detection, thus offering a promising avenue for enhancing patient outcomes and reducing mortality rates associated with skin cancer.

Moving forward, our research will focus on refining the models and exploring their applicability to other types of cancer. This will involve fine-tuning model architectures, optimizing hyperparameters, and leveraging advanced training techniques to further enhance performance and robustness. Furthermore, our future work is centered on building an automated processing pipeline to streamline the diagnostic process. Currently, the segmentation and classification models are employed sequentially, necessitating manual intervention, and hindering real-time decision-making. By integrating these components into a unified system, we aim to automate the entire workflow, from image acquisition to diagnosis, thereby improving efficiency and reducing the burden on healthcare practitioners. Additionally, we will explore the integration of real-time feedback mechanisms to enable continuous learning and adaptation of the models to evolving clinical data and practices, ensuring their relevance and effectiveness in real-world settings.

In conclusion, our study presents a state-of-the-art cancer detection framework and lays the groundwork for future advancements in oncology diagnostics. By combining innovative deep-learning techniques with clinical expertise, we have demonstrated the potential to revolutionize early cancer detection and improve patient outcomes. However, there remain several avenues for further research and development, and we are committed to addressing these challenges in our ongoing efforts to leverage artificial intelligence for the benefit of healthcare.

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